

Weighted Manifolds and Calculus on Weighted Manifolds

An Undergraduate Coursebook

A self-contained companion text for the mini-paper
on Weighted Stokes and Ornstein–Uhlenbeck Diffusion

Covering: Riemannian preliminaries, weighted measures,
weighted divergence and Laplacian, the Gaussian / OU specialization,
worked examples in finite-dimensional Hilbert spaces,
and exam-style exercises with full solutions.

Preface

This coursebook is a self-contained development of the geometric and analytic machinery needed to make sense of **weighted manifolds** and the differential operators built on them, with a deliberate eye toward the **Ornstein–Uhlenbeck operator** on Euclidean space as the central concrete example. It is written to be read alongside a draft mini-paper that treats weighted Stokes and OU diffusion, and it is intended to be a complete reference: nothing important is left to “see the literature.” The reader should be able to put this book down and write the entire foundational section of the paper from what is here.

Prerequisites. The reader is expected to be comfortable with:

- Smooth manifolds, charts, atlases, smooth maps.
- Tangent and cotangent spaces; tangent vectors as derivations.
- Differential forms, the wedge product, the exterior derivative d .
- The pullback of forms under a smooth map.
- Multivariable calculus, including the change-of-variables formula.

These are the same prerequisites the mini-paper assumes. We do **not** assume prior exposure to Riemannian geometry, Lie derivatives, or measure theory beyond the basic Lebesgue integral.

How to read this book. Chapter 1 develops just enough Riemannian geometry to define the Laplace–Beltrami operator and prove Green’s first identity. Chapter 2 introduces weighted manifolds and derives the weighted divergence operator by **two independent methods** — a measure-theoretic argument and a forms-theoretic one. The agreement of the two methods is the conceptual heart of the subject. Chapter 3 specializes everything to Euclidean space with a Gaussian weight, recovering the Ornstein–Uhlenbeck operator and its spectral theory. Chapter 4 contains six worked examples drawn from Hilbert space analysis and finite-dimensional quantum mechanics. Chapter 5 is a set of exam-style computational exercises whose full solutions appear in the appendix.

A word on conventions. Sign conventions for the Laplacian vary across the literature. Throughout this book we adopt the **analyst’s convention**: $\Delta_g f = \sum_i \partial_i^2 f$ on Euclidean $\mathbb{R}^n$, so that $-\Delta_g$ is the positive operator with nonnegative spectrum, and the heat equation reads $\partial_t u = \Delta_\mu u$. This is the convention used by the mini-paper and by most modern texts in geometric analysis and probability.

For the student who wishes to understand, not merely compute.

1 Riemannian preliminaries

A **Riemannian manifold** is a smooth manifold equipped with a smoothly varying inner product on each tangent space. This single structure is what is needed to define lengths, angles, volumes, gradients, divergences, and Laplacians intrinsically — that is, without reference to any embedding in Euclidean space. This chapter sets up the machinery and arrives at Green's first identity, the integration-by-parts theorem that powers everything in Chapter 2.

1.1 The metric tensor

Definition (Riemannian metric). Let M be a smooth n -manifold. A **Riemannian metric** on M is a smooth assignment of an inner product $g_p : T_p M \times T_p M \rightarrow \mathbb{R}$ to every point $p \in M$, such that in any chart (U, φ) with coordinates $(x^1, \dots, x^n)$, the matrix

$$g_{ij}(x) = g_p(\partial_i, \partial_j) \quad (1)$$

is symmetric and positive definite, and the entries $g_{ij}(x)$ depend smoothly on x . The pair (M, g) is called a **Riemannian manifold**. We write g^{ij} for the inverse matrix; it represents the induced inner product on the cotangent space $T_p^* M$.

The metric g is a section of the symmetric tensor bundle $\text{Sym}^2 T^* M$. Pointwise it is a symmetric bilinear form; smoothness means its component functions are smooth; positive definiteness is the condition that distinguishes a Riemannian metric from a more general (e.g., pseudo-Riemannian) one.

Example (Euclidean space). On $\mathbb{R}^n$ with standard coordinates, $g_{ij} = \delta_{ij}$ and the associated inner product is the dot product. The matrix g_{ij} is the identity in every coordinate system obtained by orthogonal transformation of $\mathbb{R}^n$, and **changes** under non-orthogonal coordinate changes (e.g., spherical coordinates).

Example (The standard Hermitian / Riemannian structure on $\mathbb{C}^m$). Identify $\mathbb{C}^m \simeq \mathbb{R}^{2m}$ via $z_k = a_k + ib_k$. The standard Hermitian inner product $\langle z, w \rangle_{\mathbb{C}^m} = \sum_k \bar{z}_k w_k$ decomposes into real and imaginary parts:

$$\langle z, w \rangle_{\mathbb{C}^m} = g(z, w) + i\omega_0(z, w), \quad (2)$$

where $g(z, w) = \text{Re} \langle z, w \rangle$ is the standard **Euclidean metric** on $\mathbb{R}^{2m}$, and $\omega_0(z, w) = \text{Im} \langle z, w \rangle$ is the standard **symplectic form**. Thus on $\mathbb{C}^m$ the metric and the symplectic form are real and imaginary halves of one Hermitian object. This decomposition is the foundation of Kähler geometry and reappears throughout the worked examples of Chapter 4.

Remark. Without a metric, one cannot speak of “length,” “orthogonality,” or “volume” on a manifold. A bare smooth manifold has only a topological structure; the metric is the **additional** structure that turns it into a geometric object. Different metrics on the same manifold yield genuinely different Riemannian manifolds.

1.2 The Riemannian volume form

Once a metric is fixed, there is a canonical way to integrate functions. The construction is uniquely determined by the requirement that a g -orthonormal cube has volume one.

Definition (Riemannian volume form). Let (M, g) be an oriented Riemannian n -manifold. The **Riemannian volume form** $d\text{vol}_g$ is the unique smooth n -form on M such that on any positively oriented g -orthonormal frame $(e_1, \dots, e_n)$ of $T_p M$,

$$d\text{vol}_g(e_1, \dots, e_n) = 1. \quad (3)$$

Proposition (Coordinate formula for the volume form). In any positively oriented chart $(U, x^1, \dots, x^n)$,

$$d\text{vol}_g = \sqrt{\det g_{ij}} dx^1 \wedge \dots \wedge dx^n. \quad (4)$$

Proof. Let $A = (A^i\{ \}_a)$ be the matrix sending the coordinate frame $(\partial_1, \dots, \partial_n)$ to a positively oriented g -orthonormal frame: $e_a = A^i\{ \}_a \partial_i$. Orthonormality reads

$$\delta_{ab} = g(e_a, e_b) = A^i\{ \}_a A^j\{ \}_b g_{ij}, \quad \text{i.e.,} \quad I = A^T G A, \quad (5)$$

where $G = (g_{ij})$. Taking determinants gives $\det A = (\det G)^{-\frac{1}{2}}$. Two top forms are related by the determinant of a change of basis, so evaluating $d\text{vol}_g$ on the coordinate frame yields

$$d\text{vol}_g(\partial_1, \dots, \partial_n) = (\det A)^{-1} = \sqrt{\det G}, \quad (6)$$

proving the coordinate formula. $\square$

Remark. The expression $\sqrt{\det g_{ij}}$ is **coordinate-dependent** — it changes value when you change charts. The form $d\text{vol}_g$ it represents is **coordinate-independent**. The two facts agree because the $\sqrt{\det g}$ factor compensates exactly for the Jacobian factor that appears when one changes coordinates inside the wedge product.

The integral of a smooth function $f : M \rightarrow \mathbb{R}$ with compact support is defined chart-by-chart: in a chart (U, x) ,

$$\int_M f d\text{vol}_g = \int_{\varphi(U)} f(x) \sqrt{\det g_{ij}(x)} dx^1 \dots dx^n, \quad (7)$$

glued via a partition of unity. Independence of the choices follows from the change-of-variables formula combined with the transformation rule for $\sqrt{\det g}$ under chart changes.

1.3 Gradient and divergence

Two operators distinguish a Riemannian manifold from a bare smooth one. Each requires the metric in an essential way.

Definition (Gradient). Let $f \in C^\infty(M)$. The **gradient** of f is the unique vector field ∇f such that

$$g(\nabla f, X) = df(X) = X(f) \quad \text{for every vector field } X. \quad (8)$$

In coordinates,

$$(\nabla f)^i = g^{ij} \partial_j f. \quad (9)$$

The differential df is a 1-form and exists on any smooth manifold. The **gradient** ∇f is a vector field, obtained from df by raising the index using g^{ij} . Without a metric, there is no canonical way to turn a covector into a vector, so there is no canonical gradient. This is the content of the often-quoted slogan “ ∇f uses a metric, df does not.”

Definition (Divergence (Lie-derivative form)). Let X be a smooth vector field on (M, g) . The **divergence** of X is the unique smooth function $\operatorname{div}_g X$ such that

$$\mathcal{L}_X(d\operatorname{vol}_g) = (\operatorname{div}_g X) d\operatorname{vol}_g, \quad (10)$$

where $\mathcal{L}_X$ denotes the Lie derivative.

Geometrically, $\operatorname{div}_g X$ measures the infinitesimal rate at which the Riemannian volume form is stretched by the flow of X . Cartan’s magic formula $\mathcal{L}_X = d \circ \iota_X + \iota_X \circ d$ together with the identity $d(d\operatorname{vol}_g) = 0$ (because $d\operatorname{vol}_g$ is a top form and so its exterior derivative is automatically zero) gives the equivalent expression

$$(\operatorname{div}_g X) d\operatorname{vol}_g = d(\iota_X d\operatorname{vol}_g), \quad (11)$$

which we will use in Chapter 2.

Proposition (Coordinate formula for divergence). In coordinates, with $X = X^i \partial_i$,

$$\operatorname{div}_g X = \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} X^i). \quad (12)$$

Proof. Compute $\iota_X d\operatorname{vol}_g$ in coordinates:

$$\iota_X d\operatorname{vol}_g = \sum_i (-1)^{i-1} X^i \sqrt{\det g} dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n, \quad (13)$$

where the hat denotes omission. Apply d and divide through by $\sqrt{\det g}$ to get the stated formula. The signs cancel because $d(\sqrt{\det g} X^i dx^1 \wedge \dots \wedge \widehat{dx^i} \wedge \dots \wedge dx^n) = (-1)^{i-1} \partial_i(\sqrt{\det g} X^i) d\text{vol}_g$.

□

1.4 The divergence theorem and Green's identity

Stokes's theorem applied to the $(n-1)$ -form $\iota_X d\text{vol}_g$ gives the **divergence theorem**: for X compactly supported on a manifold with boundary,

$$\int_M (\text{div}_g X) d\text{vol}_g = \int_{\partial M} g(X, \nu) d\sigma, \quad (14)$$

where ν is the outward unit normal and $d\sigma$ is the induced volume form on the boundary. On a closed manifold (no boundary) or for compactly supported X on a manifold without boundary, the right-hand side is zero.

Theorem (Green's first identity). Let $u, v \in C_c^\infty(M)$ be compactly supported smooth functions on (M, g) . Then

$$\int_M u \Delta_g v d\text{vol}_g = - \int_M g(\nabla u, \nabla v) d\text{vol}_g = \int_M v \Delta_g u d\text{vol}_g, \quad (15)$$

where $\Delta_g f := \text{div}_g(\nabla f)$ is the **Laplace–Beltrami operator**.

Proof. Apply the divergence theorem to $X = u \nabla v$. Using $\text{div}_g(u \nabla v) = u \text{div}_g(\nabla v) + g(\nabla u, \nabla v) = u \Delta_g v + g(\nabla u, \nabla v)$ and noting that the boundary term vanishes by compact support,

$$0 = \int_M \text{div}_g(u \nabla v) d\text{vol}_g = \int_M u \Delta_g v d\text{vol}_g + \int_M g(\nabla u, \nabla v) d\text{vol}_g. \quad (16)$$

This gives the first equality. Symmetry under $u \leftrightarrow v$ gives the second. □

The two consequences worth stating loudly:

1. **Δ_g is symmetric on $L^2(M, d\text{vol}_g)$.** This is what guarantees a real spectrum and the existence of an orthonormal eigenbasis (under appropriate completeness and ellipticity hypotheses).
2. **Coordinate formula:**

$$\Delta_g f = \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} g^{ij} \partial_j f). \quad (17)$$

On Euclidean $\mathbb{R}^n$, $g^{ij} = \delta^{ij}$ and $\sqrt{\det g} = 1$, so $\Delta_g f = \sum_i \partial_i^2 f$. This is our sign convention.

Example (Laplace–Beltrami on the round sphere S^2). In spherical coordinates (θ, ϕ) with $g = d\theta^2 + \sin^2 \theta d\phi^2$,

$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}, \quad g^{ij} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{\sin^2 \theta} \end{pmatrix}, \quad \sqrt{\det g} = \sin \theta. \quad (18)$$

The Laplace–Beltrami operator is

$$\Delta_{S^2} f = \frac{1}{\sin} \theta \partial_\theta (\sin \theta \partial_\theta f) + \frac{1}{\sin^2} \theta \partial_\phi^2 f. \quad (19)$$

The eigenvalues are $-\ell(\ell + 1)$ for $\ell = 0, 1, 2, \dots$, with multiplicity $2\ell + 1$, and the eigenfunctions are the spherical harmonics Y_ℓ^m . The total volume is $\int_{S^2} d\text{vol}_g = \int_0^\pi \int_0^{2\pi} \sin \theta d\phi d\theta = 4\pi$, as expected.

This finishes the Riemannian preliminaries. We now move to weighted manifolds, where the volume form is replaced by a more general positive measure and the operators above acquire a correction term.

2 Weighted manifolds

A **weighted manifold** is a Riemannian manifold whose Riemannian volume form has been replaced by a smooth positive measure of one's own choosing. This is the natural setting for analysis whenever the relevant measure is not the geometric one — as is the case for Gaussian measures, invariant measures of stochastic processes, equilibrium states of physical systems, and the natural inner products of quantum mechanics with confining potentials.

2.1 The weighted measure

Definition (Weighted manifold). A **weighted manifold** is a triple (M, g, μ) where (M, g) is a Riemannian manifold and μ is a smooth positive measure on M . Equivalently, on an orientable manifold,

$$d\mu = D d\text{vol}_g \quad (20)$$

for a smooth positive **density** $D \in C^\infty(M)$. We will always write $D = e^{-V}$ for a smooth **potential** $V \in C^\infty(M)$, so that

$$d\mu = e^{-V} d\text{vol}_g. \quad (21)$$

There are two equivalent perspectives on the data: (M, g, V) as a Riemannian manifold with a chosen potential, or (M, g, μ) as a Riemannian manifold with a chosen measure. We use them interchangeably.

Example (Gaussian measure on $\mathbb{R}^n$). Let $M = \mathbb{R}^n$ with the Euclidean metric $g = \delta_{ij}$ and potential $V(x) = \frac{\|x\|^2}{2}$. The associated weighted measure is

$$d\gamma(x) = (2\pi)^{-\frac{n}{2}} e^{-\frac{\|x\|^2}{2}} dx, \quad (22)$$

the standard **Gaussian probability measure** on $\mathbb{R}^n$. (The normalization constant $(2\pi)^{-\frac{n}{2}}$ is absorbed into a redefinition of V and does not affect any of the operators we will derive.)

Example (Riemannian volume as a weighted measure). Setting $V \equiv 0$ recovers $d\mu = d\text{vol}_g$ and the entire weighted theory reduces to the unweighted theory of Chapter 1. Thus the weighted setting is a strict generalization, and every formula we derive below must collapse to its unweighted analog when $V = 0$.

The point of introducing the weight is the following: when the measure of interest is $d\mu$ rather than $d\text{vol}_g$, the operator that is symmetric on $L^2(M, d\mu)$ is **not** the Laplace–Beltrami operator Δ_g , but a modification of it. Deriving this modification is the central content of the chapter.

2.2 Weighted divergence: the measure-theoretic derivation

We will derive the **weighted divergence** operator by demanding that integration by parts work cleanly against $d\mu$. Specifically, we look for an operator div_μ such that, for compactly supported smooth u and any smooth vector field X ,

$$\int_M (\operatorname{div}_\mu X) u d\mu = - \int_M g(X, \nabla u) d\mu. \quad (23)$$

The gradient $\nabla = \nabla_g$ remains the same operator from Chapter 1 (it is determined by g , not by μ). Equation 23 is the **defining property** of div_μ , and it is uniquely characterized by it because the relation must hold for every test function u .

Theorem (Weighted divergence formula). Let (M, g, μ) be a weighted manifold with $d\mu = e^{-V} d\operatorname{vol}_g$. Then

$$\operatorname{div}_\mu X = \operatorname{div}_g X - g(\nabla V, X). \quad (24)$$

Proof. Substitute $d\mu = e^{-V} d\operatorname{vol}_g$ in the right-hand side of Equation 23 and use $g(X, \nabla u) = X(u)$:

$$- \int_M g(X, \nabla u) d\mu = - \int_M X(u) e^{-V} d\operatorname{vol}_g. \quad (25)$$

We rewrite the integrand using the Leibniz rule $X(ue^{-V}) = X(u)e^{-V} - uX(V)e^{-V}$, equivalently $X(u)e^{-V} = X(ue^{-V}) + uX(V)e^{-V}$. Since $X(V) = g(X, \nabla V)$,

$$- \int_M X(u) e^{-V} d\operatorname{vol}_g = - \int_M X(ue^{-V}) d\operatorname{vol}_g - \int_M u g(X, \nabla V) e^{-V} d\operatorname{vol}_g. \quad (26)$$

The first integral is unweighted: applying Green's first identity (or equivalently the divergence theorem for the vector field $ue^{-V}X$ followed by Leibniz),

$$- \int_M X(ue^{-V}) d\operatorname{vol}_g = - \int_M g(X, \nabla(ue^{-V})) d\operatorname{vol}_g = \int_M (\operatorname{div}_g X) ue^{-V} d\operatorname{vol}_g. \quad (27)$$

Combining the two pieces,

$$- \int_M g(X, \nabla u) d\mu = \int_M [\operatorname{div}_g X - g(\nabla V, X)] u d\mu. \quad (28)$$

Comparing with Equation 23 and using that u is arbitrary compactly supported smooth, the integrands agree, giving Equation 24. $\square$

An equivalent and very useful form of Equation 24 is

$$\operatorname{div}_\mu X = e^V \operatorname{div}_g(e^{-V} X) = \frac{1}{D} \operatorname{div}_g(DX), \quad (29)$$

which displays div_μ as the unweighted divergence **conjugated** by the density $D = e^{-V}$. Both forms will be used.

2.3 Weighted divergence: the forms derivation

We now derive the same formula by an entirely independent route, using differential forms and Cartan's identity. The agreement of the two derivations is the conceptual heart of the subject: weighted divergence is simultaneously the operator that integrates by parts against $d\mu$ and the operator that measures the rate of change of the **weighted** volume form under the flow of X .

Define the **weighted volume form**

$$\Omega_\mu := e^{-V} d\text{vol}_g, \quad (30)$$

so that integration of a function f against $d\mu$ equals integration of $f\Omega_\mu$ in the form-theoretic sense.

Definition (Weighted divergence (forms)). The **weighted divergence** of a vector field X is the unique smooth function $\text{div}_\mu X$ such that

$$\mathcal{L}_X \Omega_\mu = (\text{div}_\mu X) \Omega_\mu. \quad (31)$$

Theorem (Forms derivation of div_μ). The function $\text{div}_\mu X$ defined by Equation 31 coincides with the function from Equation 24, namely

$$\text{div}_\mu X = \text{div}_g X - g(\nabla V, X). \quad (32)$$

Proof. Since Ω_μ is a top form on M , $d\Omega_\mu = 0$. Cartan's magic formula then reduces the Lie derivative to

$$\mathcal{L}_X \Omega_\mu = d(\iota_X \Omega_\mu) + \iota_X (d\Omega_\mu) = d(\iota_X \Omega_\mu). \quad (33)$$

Compute $\iota_X \Omega_\mu = e^{-V} \iota_X d\text{vol}_g$ (interior products commute with multiplication by smooth functions). Apply d using the Leibniz rule for the wedge product:

$$d(e^{-V} \iota_X d\text{vol}_g) = d(e^{-V}) \wedge \iota_X d\text{vol}_g + e^{-V} d(\iota_X d\text{vol}_g). \quad (34)$$

We compute each term.

Second term. By Equation 11 applied to the unweighted volume form, $d(\iota_X d\text{vol}_g) = (\text{div}_g X) d\text{vol}_g$, so $e^{-V} d(\iota_X d\text{vol}_g) = (\text{div}_g X) e^{-V} d\text{vol}_g = (\text{div}_g X) \Omega_\mu$.

First term. We have $d(e^{-V}) = -e^{-V} dV$. We claim

$$dV \wedge \iota_X d\text{vol}_g = (X(V)) d\text{vol}_g = g(\nabla V, X) d\text{vol}_g. \quad (35)$$

This is a pointwise identity in any frame: write $X = X^i \partial_i$ and use that $\iota_X d\text{vol}_g = \sum_i (-1)^{i-1} X^i \sqrt{\det g} dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$, so

$$\begin{aligned}
dV \wedge \iota_X d\text{vol}_g &= \left(\sum_j (\partial_j V) dx^j \right) \wedge \left(\sum_i (-1)^{i-1} X^i \sqrt{\det g} \hat{\omega}^i \right) \\
&= \left(\sum_i X^i \partial_i V \right) \sqrt{\det g} dx^1 \wedge \cdots \wedge dx^n,
\end{aligned} \tag{36}$$

where only the $j = i$ terms survive after wedging. The right side is exactly $X(V)d\text{vol}_g = g(\nabla V, X)d\text{vol}_g$. Hence

$$d(e^{-V}) \wedge \iota_X d\text{vol}_g = -g(\nabla V, X)e^{-V} d\text{vol}_g = -g(\nabla V, X)\Omega_\mu. \tag{37}$$

Combining.

$$\mathcal{L}_X \Omega_\mu = (\text{div}_g X)\Omega_\mu - g(\nabla V, X)\Omega_\mu = (\text{div}_g X - g(\nabla V, X))\Omega_\mu, \tag{38}$$

identifying $\text{div}_\mu X = \text{div}_g X - g(\nabla V, X)$. $\square$

The two derivations land on the same formula. The measure-theoretic path emphasizes the role of div_μ as the negative adjoint of ∇ on $L^2(d\mu)$; the forms path emphasizes its role as a “logarithmic derivative” of the weighted volume form along the flow of X . Both views are useful; both will be used in the worked examples.

2.4 The weighted Laplacian and weighted Green’s identity

Definition (Weighted Laplacian). Let (M, g, μ) be a weighted manifold. The **weighted Laplacian** (also known as the **Bakry–Émery Laplacian** or the **drift Laplacian**) is

$$\Delta_\mu f := \text{div}_\mu(\nabla f). \tag{39}$$

By Equation 24, this evaluates to

$$\Delta_\mu f = \Delta_g f - g(\nabla V, \nabla f). \tag{40}$$

In divergence form, using Equation 29,

$$\Delta_\mu f = \frac{1}{D} \text{div}_g(D \nabla f) = \frac{1}{e^{-V} \sqrt{\det g}} \partial_i (e^{-V} \sqrt{\det g} g^{ij} \partial_j f). \tag{41}$$

The structure parallels the unweighted formula exactly, with $\sqrt{\det g}$ replaced by $D\sqrt{\det g} = e^{-V} \sqrt{\det g}$.

Theorem (Weighted Green’s first identity). Let $u, v \in C_c^\infty(M)$. Then

$$\int_M u \Delta_\mu v d\mu = - \int_M g(\nabla u, \nabla v) d\mu = \int_M v \Delta_\mu u d\mu. \tag{42}$$

Proof. Apply Equation 23 with $X = \nabla v$:

$$\int_M (\operatorname{div}_\mu \nabla v) u d\mu = - \int_M g(\nabla v, \nabla u) d\mu, \quad (43)$$

which is the first equality. Symmetry of $g(\nabla u, \nabla v)$ under $u \leftrightarrow v$ gives the second. $\square$

The punchline: Δ_μ is symmetric on $L^2(M, d\mu)$. This is the **correct** Hilbert space for the weighted operator. Symmetry on $L^2(d\operatorname{vol}_g)$ would require a different operator (essentially the **Witten Laplacian** of Section 2.5 below). Choosing the right Hilbert space for the right operator is part of the data of the weighted theory.

Remark. Identity Equation 42 has profound consequences. Under appropriate ellipticity and completeness assumptions, Δ_μ extends to a self-adjoint operator on $L^2(M, d\mu)$ with a real, discrete spectrum and a complete orthonormal eigenbasis. The lowest eigenvalue is zero, attained by the constant function (assuming $\mu(M) < \infty$); the spectral gap controls exponential rates of convergence to equilibrium for the heat semigroup $e^{t\Delta_\mu}$. This is the geometric content of modern functional inequalities (Poincaré, log-Sobolev) and is one of the reasons weighted manifolds are studied so intensely in geometric analysis.

2.5 A second perspective: the Witten Laplacian

We pause to record an alternative operator that is **unitarily equivalent** to Δ_μ and whose form makes the connection to quantum mechanics manifest. Define the **Witten Laplacian** (with parameter $\frac{1}{2}$) by conjugation:

$$\Delta_W := -e^{-\frac{V}{2}} (-\Delta_\mu) e^{\frac{V}{2}}. \quad (44)$$

A direct computation (Exercise 2.7) gives the Schrödinger-like form

$$\Delta_W f = \Delta_g f - \frac{1}{4} \|\nabla V\|^2 f + \frac{1}{2} (\Delta_g V) f. \quad (45)$$

In particular, Δ_W acts on $L^2(M, d\operatorname{vol}_g)$ rather than $L^2(d\mu)$, and the unitary $f \mapsto e^{\frac{V}{2}} f$ is an isomorphism $L^2(d\mu) \rightarrow L^2(d\operatorname{vol}_g)$ that intertwines Δ_μ and Δ_W . We will see in Chapter 4 that, for $V(x) = \frac{\|x\|^2}{2}$ on Euclidean space, $-\Delta_W$ becomes (up to a constant) the **quantum harmonic oscillator Hamiltonian**. This is the precise mechanism by which a probabilistic object (OU diffusion) and a quantum object (harmonic oscillator) are two faces of one operator. We return to this in §4.4.

3 The Gaussian case: Ornstein–Uhlenbeck

We now specialize the entire weighted theory to the most important concrete example: Euclidean space with a Gaussian weight. The resulting weighted Laplacian is the **Ornstein–Uhlenbeck operator**, the central object of the mini-paper. Everything in this chapter is a corollary of Chapter 2; the goal is to make the corollary concrete, write down explicit spectral data, and prepare the ground for numerical verification.

3.1 Specialization

Take $M = \mathbb{R}^n$ with the Euclidean metric, so $d\text{vol}_g = dx$ and $\Delta_g = \sum_i \partial_i^2 =: \Delta$. Take the potential

$$V(x) = \frac{\|x\|^2}{2}. \quad (46)$$

The associated weighted measure is the **standard Gaussian measure**

$$d\gamma(x) = (2\pi)^{-\frac{n}{2}} e^{-\frac{\|x\|^2}{2}} dx, \quad (47)$$

which is a probability measure on $\mathbb{R}^n$.

By Equation 40 with $\nabla V = x$ and $g(\nabla V, \nabla f) = x \cdot \nabla f$, we obtain the **Ornstein–Uhlenbeck operator**:

$$\Delta_\gamma f = \Delta f - x \cdot \nabla f. \quad (48)$$

The corresponding heat equation is the **Ornstein–Uhlenbeck equation**:

$$\partial_t u(t, x) = \Delta_\gamma u(t, x), \quad u(0, x) = u_0(x). \quad (49)$$

By Theorem 2.6, Δ_γ is symmetric on $L^2(\mathbb{R}^n, d\gamma)$ and admits an orthonormal eigenbasis.

3.2 Hermite polynomials in one dimension

The eigenfunctions of Δ_γ in one dimension are the **probabilist Hermite polynomials**. We define them by the recursion

$$\text{He}_0(x) = 1, \quad \text{He}_1(x) = x, \quad \text{He}_{n+1}(x) = x\text{He}_n(x) - n\text{He}_{n-1}(x). \quad (50)$$

The first few are

$$\text{He}_0 = 1, \quad \text{He}_1 = x, \quad \text{He}_2 = x^2 - 1, \quad \text{He}_3 = x^3 - 3x, \quad \text{He}_4 = x^4 - 6x^2 + 3. \quad (51)$$

Proposition (Hermite eigenvalue equation). In one dimension,

$$\Delta_\gamma \text{He}_n = -n\text{He}_n \quad \text{for every } n \geq 0. \quad (52)$$

Proof. We verify the equation by direct computation for small n and sketch the general argument via generating functions.

Base cases. In one dimension, $\Delta_\gamma f = f'' - xf'$. Hence:

- $\Delta_\gamma \text{He}_0 = \Delta_\gamma(1) = 0 = -0 \cdot \text{He}_0$.
- $\Delta_\gamma \text{He}_1 = \Delta_\gamma(x) = 0 - x \cdot 1 = -x = -1 \cdot \text{He}_1$.
- $\Delta_\gamma \text{He}_2 = \Delta_\gamma(x^2 - 1) = 2 - x \cdot 2x = 2 - 2x^2 = -2(x^2 - 1) = -2 \cdot \text{He}_2$.
- $\Delta_\gamma \text{He}_3 = \Delta_\gamma(x^3 - 3x) = 6x - x \cdot (3x^2 - 3) = -3(x^3 - 3x) = -3 \cdot \text{He}_3$.

General case. The probabilist Hermite polynomials admit the generating-function identity

$$F(x, t) := \sum_{n=0}^{\infty} \text{He}_n(x) \frac{t^n}{n!} = e^{xt - \frac{t^2}{2}}. \quad (53)$$

A direct computation gives

$$\partial_x F = tF, \quad \partial_x^2 F = t^2 F, \quad x\partial_x F = xtF, \quad (54)$$

so that $\Delta_\gamma F = \partial_x^2 F - x\partial_x F = (t^2 - xt)F$. On the other hand, $\partial_t F = (x - t)F$, hence $-t\partial_t F = (t^2 - xt)F$. Therefore

$$\Delta_\gamma F(x, t) = -t\partial_t F(x, t) = -t \sum_{n=0}^{\infty} n \text{He}_n(x) \frac{t^{n-1}}{n!} = \sum_{n=0}^{\infty} (-n) \text{He}_n(x) \frac{t^n}{n!}. \quad (55)$$

Comparing coefficients of t^n on both sides yields $\Delta_\gamma \text{He}_n = -n \text{He}_n$, as claimed. $\square$

Proposition (Orthogonality and completeness). The probabilist Hermite polynomials are orthogonal on $L^2(\mathbb{R}, d\gamma_1)$:

$$\int_{\mathbb{R}} \text{He}_m(x) \text{He}_n(x) d\gamma_1(x) = n! \delta_{mn}. \quad (56)$$

Normalized polynomials $\tilde{\text{He}}_n := \frac{\text{He}_n}{\sqrt{n!}}$ form a complete orthonormal basis of $L^2(\mathbb{R}, d\gamma_1)$.

We omit the proof; orthogonality follows from successive integration by parts using Equation 42 and induction. Completeness is a consequence of density of polynomials in $L^2(d\gamma_1)$, which itself follows from the existence of all moments of the Gaussian.

3.3 The multi-dimensional case

In dimension n , the eigenfunctions of Δ_γ on $\mathbb{R}^n$ are **tensor products** of one-dimensional Hermite polynomials. Given a multi-index $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$, define

$$\text{He}_\alpha(x) := \prod_{i=1}^n \text{He}_{\alpha_i}(x_i). \quad (57)$$

Then

$$\Delta_\gamma \text{He}_\alpha = -|\alpha| \text{He}_\alpha, \quad \text{where } |\alpha| = \sum_i \alpha_i. \quad (58)$$

The eigenvalue $-k$ has multiplicity equal to the number of multi-indices $\alpha \in \mathbb{N}^n$ with $|\alpha| = k$, namely $\binom{n+k-1}{k}$. The tensor products $\tilde{\text{He}}_\alpha = \prod_i \frac{\text{He}_{\alpha_i}}{\sqrt{\alpha_i!}}$ form a complete orthonormal basis of $L^2(\mathbb{R}^n, d\gamma_n)$, where $d\gamma_n$ is the standard Gaussian on $\mathbb{R}^n$.

3.4 The OU semigroup and Mehler's formula

The solution of the OU heat equation Equation 49 with initial condition u_0 admits an explicit integral representation. Define the **Ornstein–Uhlenbeck semigroup** by $P_t := e^{t\Delta_\gamma}$. Then for $t > 0$,

$$(P_t f)(x) = \int_{\mathbb{R}^n} f(e^{-t}x + \sqrt{1 - e^{-2t}}y) d\gamma_n(y). \quad (59)$$

This is **Mehler's formula**. Equivalently, P_t has the integral kernel

$$K_t(x, y) = \frac{1}{(1 - e^{-2t})^{\frac{n}{2}}} \exp\left(-\frac{e^{-2t}(\|x\|^2 + \|y\|^2) - 2e^{-t}x \cdot y}{2(1 - e^{-2t})}\right). \quad (60)$$

Two important consequences:

1. **Eigenmode decay.** On Hermite eigenfunctions, $P_t \text{He}_\alpha = e^{-|\alpha|t} \text{He}_\alpha$. The lowest mode ($|\alpha| = 0$, the constant) is preserved; higher modes decay exponentially with rate equal to the eigenvalue magnitude.
2. **Convergence to equilibrium.** As $t \rightarrow \infty$, $(P_t f)(x) \rightarrow \int f d\gamma_n$ for every $f \in L^2(d\gamma_n)$, and the convergence is exponential in $L^2(d\gamma_n)$ with rate at least 1 (the **spectral gap**).

These two facts will serve as the analytical benchmarks for the C++ verification described in the mini-paper. Equilibrium is the constant function (the Gaussian moments of the initial datum), and the convergence rate towards it is the spectral gap.

Example (OU on $\mathbb{R}^2$: three lowest eigenvalues). In dimension $n = 2$, multi-indices and eigenvalues are:

$$\begin{aligned} \alpha = (0, 0) : \quad \lambda &= 0, \quad \text{He}_{0,0} = 1, \quad \text{multiplicity } 1. \\ \alpha \in \{(1, 0), (0, 1)\} : \quad \lambda &= -1, \quad \text{spanned by } x_1, x_2, \quad \text{multiplicity } 2. \\ \alpha \in \{(2, 0), (1, 1), (0, 2)\} : \quad \lambda &= -2, \quad \text{spanned by } x_1^2 - 1, x_1 x_2, x_2^2 - 1, \quad \text{multiplicity } 3. \end{aligned} \quad (61)$$

In general $\lambda = -k$ has multiplicity $k + 1$ in two dimensions.

This completes the foundational theory. We now turn to extended worked examples that connect the weighted-manifold formalism to Hilbert spaces and finite-dimensional quantum mechanics.

4 Worked examples

The examples below are arranged from purely analytic (Examples 4.1–4.2), through Hilbert-space and Fourier-theoretic settings (Examples 4.3–4.4), to finite-dimensional quantum mechanics in dimensions 3, 4, and general n (Examples 4.5–4.6). Each example is fully worked.

4.1 Example 4.1 — The harmonic oscillator and the OU operator

Set $M = \mathbb{R}$ with Euclidean metric and $V(x) = \frac{x^2}{2}$. The OU operator is

$$\Delta_\gamma f = f'' - x f'. \quad (62)$$

Define the **unitary** multiplication operator $U : L^2(\mathbb{R}, d\gamma_1) \rightarrow L^2(\mathbb{R}, dx)$ by $(Uf)(x) = e^{-\frac{x^2}{4}} f(x)$. The fact that U is unitary follows from the change of measure: $\int |(Uf)(x)|^2 dx = \int |f(x)|^2 e^{-\frac{x^2}{2}} dx = \int |f(x)|^2 d\gamma_1(x)$. Conjugating $-\Delta_\gamma$ by U yields a Schrödinger operator. We compute. For $g \in C_c^\infty(\mathbb{R})$, set $f = U^{-1}g = e^{\frac{x^2}{4}}g$. Then

$$\begin{aligned} f' &= \left(\frac{x}{2}\right) e^{\frac{x^2}{4}} g + e^{\frac{x^2}{4}} g' = e^{\frac{x^2}{4}} \left(g' + \left(\frac{x}{2}\right)g\right), \\ f'' &= e^{\frac{x^2}{4}} \left(g'' + xg' + \left(\frac{1}{2} + \frac{x^2}{4}\right)g\right). \end{aligned} \quad (63)$$

Hence

$$\Delta_\gamma f = f'' - x f' = e^{\frac{x^2}{4}} \left(g'' + xg' + \left(\frac{1}{2} + \frac{x^2}{4}\right)g\right) - x e^{\frac{x^2}{4}} \left(g' + \left(\frac{x}{2}\right)g\right), \quad (64)$$

and the cross terms cancel:

$$\Delta_\gamma f = e^{\frac{x^2}{4}} \left(g'' + \left(\frac{1}{2} - \frac{x^2}{4}\right)g\right). \quad (65)$$

Therefore $U\Delta_\gamma U^{-1}g = e^{-\frac{x^2}{4}}\Delta_\gamma f = g'' + \left(\frac{1}{2} - \frac{x^2}{4}\right)g$, or equivalently

$$-U\Delta_\gamma U^{-1} = -g'' + \left(\frac{x^2}{4} - \frac{1}{2}\right)g. \quad (66)$$

This is, up to constants, the **quantum harmonic oscillator Hamiltonian** $H = -\frac{1}{2}\frac{d^2}{dx^2}x^2 + \frac{1}{2}x^2$ (specifically, $-U\Delta_\gamma U^{-1} = 2H - 1$ after the rescaling that matches conventions). The eigenvalues $-n$ of Δ_γ correspond, after this unitary, to the oscillator eigenvalues $E_n = n + \frac{1}{2}$. The mini-paper's OU dynamics is therefore **imaginary-time Schrödinger evolution** under the harmonic oscillator Hamiltonian, viewed in the unitary frame where the ground state has been factored out.

4.2 Example 4.2 — A non-Gaussian weight and a tilted Laplacian

Let $M = \mathbb{R}$, Euclidean metric, $V(x) = \frac{x^4}{4}$. Then $\nabla V = x^3$ and

$$\Delta_\mu f = f'' - x^3 f'. \quad (67)$$

Although the analogous Hermite theory is no longer polynomial-friendly, the **defining identity** Equation 23 still gives integration by parts: for compactly supported u, v ,

$$\int_{\mathbb{R}} u(x)(v''(x) - x^3 v'(x))e^{-\frac{x^4}{4}} dx = - \int_{\mathbb{R}} u'(x)v'(x)e^{-\frac{x^4}{4}} dx. \quad (68)$$

The lowest eigenvalue is again 0 (constant function); higher eigenvalues are no longer integers and must be computed numerically. This example shows that the entire weighted formalism is robust under the choice of V : only when V is quadratic do the eigenfunctions become polynomial (Hermite). For other potentials, weighted Green's identity still holds and remains the right tool for spectral analysis.

4.3 Example 4.3 — Finite Fourier basis as a weighted Hilbert space

Take the finite Fourier space

$$H_N = \text{span}_{\mathbb{C}}\{e_{-N}, \dots, e_N\} \subset L^2(S^1; \mathbb{C}), \quad e_k(x) = (2\pi)^{-\frac{1}{2}} e^{ikx}. \quad (69)$$

Identify $H_N \simeq \mathbb{C}^{2N+1}$ via $u = \sum_{k=-N}^N z_k e_k$. The L^2 inner product on S^1 reduces, by orthonormality of the Fourier basis, to the Hermitian inner product on $\mathbb{C}^{2N+1}$:

$$\langle u, v \rangle_{L^2} = \sum_{k=-N}^N \overline{z_k} w_k. \quad (70)$$

Now treat $\mathbb{C}^{2N+1}$ as a real $2(2N+1)$ -dimensional manifold with the Euclidean metric (Example 1.2) and consider the Gaussian weight $d\mu(z) = (2\pi)^{-(2N+1)} e^{-\|z\|^2} dz$, where the norm and volume measure are real-Euclidean. The associated weighted Laplacian Δ_μ acts on functions $f : H_N \rightarrow \mathbb{R}$ as

$$\Delta_\mu f = \sum_{k=-N}^N (\partial_{a_k}^2 + \partial_{b_k}^2) f - 2 \sum_k (a_k \partial_{a_k} + b_k \partial_{b_k}) f, \quad (71)$$

where $z_k = a_k + ib_k$. The factor 2 comes from $V(z) = \|z\|^2$ (no factor of $\frac{1}{2}$) so $\nabla V = 2z$ in real coordinates. This is the $2(2N+1)$ -dimensional OU operator. Its eigenfunctions are the real-coordinate Hermite polynomials, and its lowest nonzero eigenvalue is -2 (one factor 1 from each pair (a_k, b_k) but all rescaled by 2).

This example shows how the finite Fourier space, which already played a central role in the mini-paper's $\mathbb{CP}^{m-1}$ construction, **also** carries a weighted-manifold structure naturally compatible with Gaussian noise. In quantum information language, this is the setup for **Gaussian states** and their time evolution.

4.4 Example 4.4 — Imaginary-time Schrödinger as OU on a finite Hilbert space

Consider the Hilbert space $H = \mathbb{C}^n$ (a “qudit” of dimension n) with its standard Hermitian inner product, and a self-adjoint Hamiltonian $H_0 : \mathbb{C}^n \rightarrow \mathbb{C}^n$. The **imaginary-time Schrödinger equation**

$$\partial_\tau \psi = -H_0 \psi \quad (72)$$

on $\mathbb{C}^n$ describes the projection of an arbitrary initial state onto the ground state of H_0 (assuming the ground state is unique and the spectrum is bounded below): writing $\psi(0) = \sum_k c_k \phi_k$ in the eigenbasis of H_0 with eigenvalues E_k , the solution is

$$\psi(\tau) = \sum_k c_k e^{-\tau E_k} \phi_k, \quad (73)$$

which converges (after normalization) to the lowest-energy eigenstate as $\tau \rightarrow \infty$. This is exactly the structure of the OU heat equation with the lowest Hermite mode (the constant) playing the role of the ground state. In the harmonic-oscillator case (Example 4.1), the two constructions are **unitarily equivalent**. In the general qudit case, both are instances of the same abstract pattern: a weighted heat equation on an appropriate measure space.

4.5 Example 4.5 — A qutrit ($n = 3$) and the projective space $\mathbb{CP}^2$

Let $H = \mathbb{C}^3$ with standard inner product. Pure quantum states are elements of the unit sphere

$$\mathcal{S}_3 = \{z \in \mathbb{C}^3 : \|z\| = 1\} \subset \mathbb{C}^3, \quad (74)$$

modulo global phase; the manifold of physical pure states is

$$\mathbb{CP}^2 = \frac{\mathcal{S}_3}{S^1}. \quad (75)$$

This is the example we developed in the mini-paper at the level of the Fubini–Study form ω_{FS} . Here we examine the **Riemannian** and **weighted** structure.

The Fubini–Study **metric** (the Hermitian counterpart of the symplectic form ω_{FS}) is, in the affine chart $w = (w_1, w_2)$ on $U_0 = \{[1 : w_1 : w_2]\}$,

$$g_{j\bar{\ell}} = \frac{(1 + \|w\|^2) \delta_{j\ell} - \bar{w}_j w_\ell}{(1 + \|w\|^2)^2}. \quad (76)$$

This is the Hermitian metric on $\mathbb{CP}^2$ whose Kähler form is $\omega_{FS} = i\partial\bar{\partial} \log(1 + \|w\|^2)$. The **real** Riemannian metric is twice the real part of $g_{j\bar{\ell}}$, and its volume form is the top and power $\frac{\omega_{FS}^2}{2!}$ from the mini-paper.

Now introduce a weight. A **thermal state** of a Hamiltonian $\hat{H} : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ at inverse temperature β corresponds, on $\mathbb{CP}^2$, to the weighted measure

$$d\mu_\beta = e^{-\beta \langle z, \hat{H} z \rangle} d\text{vol}_{FS} \quad (77)$$

where $\langle z, \hat{H}z \rangle$ is the **energy expectation** function on the unit sphere, descended to $\mathbb{CP}^2$. (This descends because energy expectation is S^1 -invariant: $\langle e^{i\theta}z, \hat{H}e^{i\theta}z \rangle = \langle z, \hat{H}z \rangle$.) The associated weighted Laplacian Δ_{μ_β} governs **imaginary-time evolution of distributions over quantum states** under the Hamiltonian $\hat{H}$. The eigenfunctions of Δ_{μ_β} on $\mathbb{CP}^2$ are not classical objects: they encode information about how a probability distribution over pure qutrit states relaxes to thermal equilibrium. This is the geometric setting underlying **quantum metrology** and **Bayesian quantum estimation**.

The specific calculation of Δ_{μ_β} on $\mathbb{CP}^2$ is left as Exercise 5.16. The qutrit is the smallest dimension where the geometry is nontrivial: $\mathbb{CP}^1 \simeq S^2$ is the Bloch sphere, but $\mathbb{CP}^2$ is the first projective space whose Riemannian curvature is genuinely multi-dimensional.

4.6 Example 4.6 — Two qubits ($n = 4$) and a Gaussian weight on coherent states

A two-qubit system has Hilbert space $H = \mathbb{C}^4 = \mathbb{C}^2 \otimes \mathbb{C}^2$. Pure states form $\frac{\mathcal{S}_4}{S^1} = \mathbb{CP}^3$. A **separable** (product) state is one of the form $z = z^A \otimes z^B$ with $z^A, z^B \in \mathcal{S}_2$; the set of separable pure states is

$$\text{Sep} \simeq \mathbb{CP}^1 \times \mathbb{CP}^1 \subset \mathbb{CP}^3. \quad (78)$$

The complement is the set of **entangled** pure states. The Fubini–Study metric on $\mathbb{CP}^3$ restricts to a metric on Sep , which by a direct computation (Exercise 5.17) equals the **product** of the Fubini–Study metrics on the two $\mathbb{CP}^1$ factors.

Now place on $\mathcal{S}_4$ the Gaussian weight inherited from $\mathbb{C}^4$, i.e., $d\mu = e^{-\|z\|^2} dz$ on $\mathbb{C}^4$, restricted (or projected) to the sphere. The associated weighted Laplacian on $\mathcal{S}_4$ has a kernel along the phase direction (as in the mini-paper’s analysis of ω_0 restricted to the sphere) and descends to a weighted Laplacian on $\mathbb{CP}^3$. Its lowest eigenvalues, computed explicitly, control the relaxation rate of distributions of two-qubit pure states under thermal fluctuations. The fact that the separable and entangled subsets have **different** Riemannian and weighted geometries (separable being a codimension-2 submanifold) is the geometric underpinning of why entanglement is a distinguished resource.

4.7 Example 4.7 — A general n -dimensional qudit

Generalize to $H = \mathbb{C}^n$. The manifold of pure states is $\mathbb{CP}^{n-1}$ of real dimension $2(n-1)$. The Fubini–Study metric in any affine chart $w = (w_1, \dots, w_{n-1})$ on U_0 is

$$g_{j\bar{\ell}} = \frac{(1 + \|w\|^2)\delta_{j\ell} - \overline{w_j}w_\ell}{(1 + \|w\|^2)^2}. \quad (79)$$

For a Hamiltonian $\hat{H} : \mathbb{C}^n \rightarrow \mathbb{C}^n$ with eigenvalues $E_1 \leq \dots \leq E_n$ and eigenvectors ϕ_k , the energy expectation function $E(z) = \langle z, \hat{H}z \rangle_{\|z\|^2}$ descends to a smooth function on $\mathbb{CP}^{n-1}$ whose critical points are exactly the eigenstates $[\phi_k]$, with Morse index $2(k-1)$ (i.e., 0 for the ground state, $2(n-1)$ for the highest excited state). This is a classical theorem of Atiyah–Bott, and it is the simplest case of the **Morse theory of the Fubini–Study energy function**.

The weighted Laplacian Δ_{μ_β} on $\mathbb{CP}^{n-1}$ for the thermal weight $d\mu_\beta = e^{-\beta E} d\text{vol}_{FS}$ is then a self-adjoint operator on $L^2(\mathbb{CP}^{n-1}, d\mu_\beta)$, and its low-lying spectrum encodes the **quantum information geometry** of thermal qudit states. As $\beta \rightarrow \infty$, the measure μ_β concentrates on the ground state $[\phi_1]$, and the spectral gap of Δ_{μ_β} controls how quickly distributions relax to this ground-state delta. As $\beta \rightarrow 0$, μ_β becomes uniform Fubini–Study measure, and one recovers the spectrum of the **unweighted** Laplacian on $\mathbb{CP}^{n-1}$, which has eigenvalues $-k(k+n-1)$ for $k = 0, 1, 2, \dots$ (the projective-space analog of the spherical-harmonic spectrum on S^2).

These are the natural generalizations of the OU operator from Chapter 3 to the geometric setting of finite-dimensional quantum systems. The exercises develop the relevant computations in detail.

5 Exam-style exercises

Exercises are graded by difficulty: $\star$ (warm-up), $\star$ (standard), $\star \star$ (challenging). Full solutions to every exercise are given in the appendix.

5.1 Riemannian preliminaries

Exercise 1.1. ($\star$) On $\mathbb{R}^2$ with polar coordinates (r, θ) , the Euclidean metric is $g = dr^2 + r^2 d\theta^2$. Write down g_{ij} , g^{ij} , $\sqrt{\det g}$, and the Laplace–Beltrami operator Δ_g in these coordinates.

Exercise 1.2. ($\star$) On $\mathbb{R}^3$ with spherical coordinates (r, θ, ϕ) and metric $g = dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2$, compute $\sqrt{\det g}$ and write down the Laplace–Beltrami operator.

Exercise 1.3. ($\star$) Verify Green’s first identity directly for $M = \mathbb{R}^3$, $u(x) = e^{-\|x\|^2}$, $v(x) = \|x\|^2$, by computing both sides of $\int u \Delta v dx = - \int \nabla u \cdot \nabla v dx$.

Exercise 1.4. ($\star$) On the round 2-sphere S^2 with the metric of §1.4, show by direct computation that the function $f(\theta, \phi) = \cos \theta$ is an eigenfunction of Δ_{S^2} with eigenvalue -2 .

5.2 Weighted manifolds

Exercise 2.1. ($\star$) For $M = \mathbb{R}$ with Euclidean metric and $V(x) = \alpha x$ (linear potential, $\alpha \in \mathbb{R}$), compute Δ_μ explicitly.

Exercise 2.2. ($\star$) Verify the conjugation formula $\operatorname{div}_\mu X = D^{-1} \operatorname{div}_g (DX)$ when $D = e^{-V}$ by direct expansion of the right-hand side using the unweighted divergence formula.

Exercise 2.3. ($\star$) Let $M = \mathbb{R}$, $V(x) = \frac{x^2}{2}$. Verify weighted Green’s identity for $u(x) = x$, $v(x) = x^2 - 1$ by explicitly computing both sides of $\int u \Delta_\mu v d\gamma_1 = - \int u' v' d\gamma_1$.

Exercise 2.4. ($\star$) Show that on a weighted manifold with $d\mu = e^{-V} d\operatorname{vol}_g$, the conjugated operator $\Delta_W := -e^{-\frac{V}{2}} (-\Delta_\mu) e^{\frac{V}{2}}$ has the Schrödinger form

$$\Delta_W f = \Delta_g f - \frac{1}{4} \|\nabla V\|_g^2 f + \frac{1}{2} (\Delta_g V) f. \quad (80)$$

(Δ_W is the **Witten Laplacian**.)

Exercise 2.5. (★) Show that for a smooth vector field X on a weighted manifold,

$$\operatorname{div}_\mu X = \operatorname{div}_{c\mu} X \quad (81)$$

for any constant $c > 0$. That is, the weighted divergence depends only on the **measure class**, not on the overall scale of μ .

Exercise 2.6. (★ ★) A **gradient flow** of a function $F : M \rightarrow \mathbb{R}$ on a weighted manifold is a curve $\gamma(t)$ satisfying $\dot{\gamma}(t) = -\nabla F(\gamma(t))$. Show that along such a flow,

$$\frac{d}{dt} F(\gamma(t)) = -\|\nabla F\|^2 \leq 0, \quad (82)$$

so F is non-increasing. Then show that the **evolution equation** for the density $\rho(t, x)$ of an ensemble of particles undergoing this flow in the presence of weighted Brownian noise (the **Fokker–Planck equation**) is

$$\partial_t \rho = \operatorname{div}_\mu(\rho \nabla F). \quad (83)$$

Identify this with the weighted heat equation when $F = -V$.

5.3 The Gaussian / OU case

Exercise 3.1. (★) Verify that $\operatorname{He}_4(x) = x^4 - 6x^2 + 3$ satisfies $\Delta_\gamma \operatorname{He}_4 = -4\operatorname{He}_4$.

Exercise 3.2. (★) Compute $\int_{\mathbb{R}} \operatorname{He}_2(x)^2 d\gamma_1(x)$ and confirm it equals $2! = 2$.

Exercise 3.3. (★) Solve the OU heat equation on $\mathbb{R}$ with initial datum $u_0(x) = x^2$. (Hint: write u_0 in the Hermite basis.)

Exercise 3.4. (★) Solve the OU heat equation on $\mathbb{R}$ with initial datum $u_0(x) = e^{-\frac{x^2}{2}}$ using Mehler's formula. Express the answer in closed form.

Exercise 3.5. (★) On $\mathbb{R}^2$ with the standard Gaussian, list all eigenfunctions of Δ_γ with eigenvalue -3 and verify orthogonality of two of them.

5.4 Hilbert-space and quantum settings

Exercise 4.1. (★) On $\mathbb{C}^2 \simeq \mathbb{R}^4$ with Euclidean metric and $V(z) = \|z\|^2$, compute Δ_μ in real coordinates (a_1, b_1, a_2, b_2) and verify that all six bilinear monomials $a_1 a_2 - b_1 b_2$, $a_1 b_2 + a_2 b_1$, $a_1^2 - b_1^2$, etc., are eigenfunctions. Identify the eigenvalues.

Exercise 4.2. (★) On S^2 (the qubit Bloch sphere $\mathbb{CP}^1$), the Fubini–Study metric is, up to a constant, the round metric. Compute the lowest eigenvalues of the unweighted Laplace–Beltrami operator.

Exercise 4.3. (★ ★) Let $\hat{H} = \text{diag}(0, 1, 2)$ on $\mathbb{C}^3$. The energy expectation function $E([z]) = \sum_k k|z_k|^2$ on $\mathbb{CP}^2$ is a smooth function. Find its critical points and determine the Hessian at each one. Verify the Atiyah–Bott Morse-index claim (Example 4.7).

Exercise 4.4. (★ ★) Let $\beta > 0$ and consider the thermal weight $d\mu_\beta = e^{-\beta E} d\text{vol}_{FS}$ on $\mathbb{CP}^2$ with E as in the previous exercise. Show that as $\beta \rightarrow \infty$, the measure $\frac{\mu_\beta}{\mu_\beta}(\mathbb{CP}^2)$ concentrates on the ground state $[1 : 0 : 0]$, and estimate the rate of concentration.

Exercise 4.5. (★ ★) On the qutrit projective space $\mathbb{CP}^2$ in the affine chart (w_1, w_2) , write out the weighted Laplacian Δ_{μ_β} in coordinates, where E is given as in Exercise 4.3. (You will need both the Fubini–Study metric components from §4.5 and the formula $\Delta_\mu = \Delta_g - g(\nabla V, \nabla \cdot)$.)

Exercise 4.6. (★ ★) Show that on the two-qubit space $\mathbb{CP}^3$, the separable submanifold $\text{Sep} \simeq \mathbb{CP}^1 \times \mathbb{CP}^1$ inherits the **product** Fubini–Study metric from the ambient $\mathbb{CP}^3$. (Hint: use the embedding $(z^A, z^B) \mapsto z^A \otimes z^B$ and pull back the Fubini–Study Hermitian form.)

6 Solutions to selected exercises

We give complete solutions to the exercises of Chapter 5. The reader is strongly urged to attempt the problem before consulting the solution.

6.1 Solutions for §5.1

Solution to Exercise 1.1. In polar coordinates,

$$g_{ij} = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}, \quad g^{ij} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{r^2} \end{pmatrix}, \quad \sqrt{\det g} = r. \quad (84)$$

The Laplace–Beltrami operator is

$$\Delta_g f = \frac{1}{r} \partial_r (r \partial_r f) + \frac{1}{r^2} \partial_\theta^2 f = \partial_r^2 f + \frac{1}{r} \partial_r f + \frac{1}{r^2} \partial_\theta^2 f. \quad (85)$$

Solution to Exercise 1.2. In spherical coordinates,

$$\sqrt{\det g} = r^2 \sin \theta, \quad g^{ij} = \text{diag} \left(1, \frac{1}{r^2}, \frac{1}{r^2 \sin^2 \theta} \right). \quad (86)$$

Hence

$$\Delta_g f = \frac{1}{r^2} \partial_r (r^2 \partial_r f) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta f) + \frac{1}{r^2 \sin^2 \theta} \partial_\phi^2 f. \quad (87)$$

Solution to Exercise 1.3. With $u = e^{-r^2}$, $v = r^2$ in $\mathbb{R}^3$ (writing $r = \|x\|$): $\Delta v = 2 + 2 + 2 = 6$, so $\int u \Delta v dx = 6 \int e^{-r^2} dx = 6\pi^{\frac{3}{2}}$. $\nabla u = -2xe^{-r^2}$, $\nabla v = 2x$, so $\nabla u \cdot \nabla v = -4r^2 e^{-r^2}$, giving $-\int \nabla u \cdot \nabla v dx = 4 \int r^2 e^{-r^2} dx = 4 \cdot \left(\frac{3}{2}\right) \pi^{\frac{3}{2}} = 6\pi^{\frac{3}{2}}$. ✓

Solution to Exercise 1.4. $f = \cos \theta$. Then $\partial_\phi f = 0$ and $\partial_\theta f = -\sin \theta$, so $\frac{1}{\sin \theta} \partial_\theta (\sin \theta \cdot (-\sin \theta)) = -\frac{1}{\sin \theta} \partial_\theta (\sin^2 \theta) = -\frac{1}{\sin \theta} \cdot 2 \sin \theta \cos \theta = -2 \cos \theta = -2f$.

6.2 Solutions for §5.2

Solution to Exercise 2.1. $V(x) = \alpha x$, so $V'(x) = \alpha$. Hence $\Delta_\mu f = f'' - \alpha f'$. This is OU with constant drift α instead of linear drift x .

Solution to Exercise 2.2. Compute

$$\begin{aligned} D^{-1} \text{div}_g (DX) &= D^{-1} [(DX)^i_{,i} \text{ (coordinate version)}] \\ &= D^{-1} [\partial_i D \cdot X^i + D \cdot \partial_i X^i + \text{Christoffel-style } \sqrt{\det g} \text{ factors}]. \end{aligned} \quad (88)$$

A cleaner way: in coordinates, $D^{-1} \text{div}_g (DX) = D^{-1} \cdot \frac{1}{\sqrt{\det g}} \partial_i (D \sqrt{\det g} X^i)$. Expand $\partial_i (D \sqrt{\det g} X^i) = (\partial_i D) \sqrt{\det g} X^i + D \partial_i (\sqrt{\det g} X^i)$. Then

$$\begin{aligned}
D^{-1}\operatorname{div}_g(DX) &= \frac{\partial_i D}{D} X^i + \frac{1}{\sqrt{\det g}} \partial_i (\sqrt{\det g} X^i) \\
&= -X^i \partial_i V + \operatorname{div}_g X = \operatorname{div}_g X - g(\nabla V, X) = \operatorname{div}_\mu X. \checkmark
\end{aligned} \tag{89}$$

Solution to Exercise 2.3. $V = \frac{x^2}{2}$, so $\Delta_\mu v = v'' - xv'$. With $v = x^2 - 1$, $v' = 2x$, $v'' = 2$: $\Delta_\mu v = 2 - 2x^2 = -2(x^2 - 1)$. Then $\int u \Delta_\mu v d\gamma = -2 \int_{\mathbb{R}} x(x^2 - 1) d\gamma_1 = -2(E[x^3] - E[x]) = 0$ (both moments of an even-symmetric measure with mean zero). The right-hand side: $u' = 1$, $v' = 2x$, so $-\int u' v' d\gamma_1 = -2 \int x d\gamma_1 = 0$. $\checkmark$

Solution to Exercise 2.4. Set $f = e^{\frac{V}{2}} g$. Then $\nabla f = e^{\frac{V}{2}} (\nabla g + (\frac{1}{2})g \nabla V)$ and

$$\Delta_g f = e^{\frac{V}{2}} \left(\Delta_g g + \nabla V \cdot \nabla g + \frac{1}{2} g \Delta_g V + \frac{1}{4} \|\nabla V\|^2 g \right). \tag{90}$$

Substituting into $\Delta_\mu f = \Delta_g f - g(\nabla V, \nabla f)$ and expanding $g(\nabla V, \nabla f) = e^{\frac{V}{2}} (g(\nabla V, \nabla g) + (\frac{1}{2}) \|\nabla V\|^2 g)$:

$$\begin{aligned}
\Delta_\mu f &= e^{\frac{V}{2}} \left(\Delta_g g + \frac{1}{2} g \Delta_g V + \frac{1}{4} \|\nabla V\|^2 g - \frac{1}{2} \|\nabla V\|^2 g \right) \\
&= e^{\frac{V}{2}} \left(\Delta_g g + \frac{1}{2} g \Delta_g V - \frac{1}{4} \|\nabla V\|^2 g \right).
\end{aligned} \tag{91}$$

Hence $e^{-\frac{V}{2}} \Delta_\mu (e^{\frac{V}{2}} g) = \Delta_g g + \frac{1}{2} (\Delta_g V) g - \frac{1}{4} \|\nabla V\|^2 g$, and the sign convention $\Delta_W = -e^{-\frac{V}{2}} (-\Delta_\mu) e^{\frac{V}{2}}$ gives $\Delta_W g = \Delta_g g - \frac{1}{4} \|\nabla V\|^2 g + \frac{1}{2} (\Delta_g V) g$. $\checkmark$

Solution to Exercise 2.5. Write $c\mu = ce^{-V} d\operatorname{vol}_g = e^{-(V - \log c)} d\operatorname{vol}_g$. The new potential is $V' = V - \log c$. Then $\nabla V' = \nabla V$ (a constant shift in V has zero gradient), so $\operatorname{div}_{c\mu} X = \operatorname{div}_g X - g(\nabla V', X) = \operatorname{div}_g X - g(\nabla V, X) = \operatorname{div}_\mu X$. $\checkmark$

Solution to Exercise 2.6. Along $\dot{\gamma} = -\nabla F$, $\frac{d}{dt} F(\gamma) = \nabla F \cdot \dot{\gamma} = -\|\nabla F\|^2 \leq 0$. For the Fokker-Planck equation: a particle ensemble undergoing the gradient flow plus diffusion with respect to $d\mu$ satisfies a continuity equation $\partial_t \rho + \operatorname{div}_\mu(\rho v) = 0$ where $v = -\nabla F + \text{diffusion}$. Working out the second-order term gives $\partial_t \rho = \operatorname{div}_\mu(\rho \nabla F) + \Delta_\mu \rho$, or, when only the deterministic flow is considered, $\partial_t \rho = \operatorname{div}_\mu(\rho \nabla F)$. With $F = -V$, $\nabla F = -\nabla V$, and $\operatorname{div}_\mu(\rho \nabla F) = \operatorname{div}_\mu(-\rho \nabla V) = -\operatorname{div}_\mu(\rho \nabla V)$; expanding using $\operatorname{div}_\mu(\rho X) = \rho \operatorname{div}_\mu X + g(\nabla \rho, X)$ and the formula $\operatorname{div}_\mu(\nabla V) = \Delta_g V - \|\nabla V\|^2$ (Exercise: verify), one recovers $\partial_t \rho = \Delta_\mu \rho$ in the equilibrium-measure case.

6.3 Solutions for §5.3

Solution to Exercise 3.1. $\operatorname{He}_4 = x^4 - 6x^2 + 3$. $\operatorname{He}_{4'} = 4x^3 - 12x$, $\operatorname{He}_{4''} = 12x^2 - 12$. Hence $\Delta_\gamma \operatorname{He}_4 = 12x^2 - 12 - x(4x^3 - 12x) = 12x^2 - 12 - 4x^4 + 12x^2 = -4x^4 + 24x^2 - 12 = -4(x^4 - 6x^2 + 3) = -4\operatorname{He}_4$. $\checkmark$

Solution to Exercise 3.2.

$$\int_{\mathbb{R}} (x^2 - 1)^2 d\gamma_1 = E[(X^2 - 1)^2] = E[X^4] - 2E[X^2] + 1 = 3 - 2 + 1 = 2 = 2!. \checkmark \tag{92}$$

(Used the standard Gaussian moments $E[X^2] = 1$, $E[X^4] = 3$.)

Solution to Exercise 3.3. $u_0(x) = x^2 = \text{He}_2(x) + 1 = \text{He}_2 + \text{He}_0$. Hermite eigenvalues: $\text{He}_0 \rightarrow 0$, $\text{He}_2 \rightarrow -2$. Hence $u(t, x) = \text{He}_0 + e^{-2t}\text{He}_2 = 1 + e^{-2t}(x^2 - 1) = 1 - e^{-2t} + e^{-2t}x^2$.

Solution to Exercise 3.4. By Mehler's formula, $(P_t f)(x) = \int f(e^{-t}x + \sqrt{1 - e^{-2t}}y) d\gamma_1(y)$ with $f(y) = e^{-\frac{y^2}{2}}$. Plug in:

$$(P_t f)(x) = \int_{\mathbb{R}} e^{-\frac{(e^{-t}x + \sqrt{1 - e^{-2t}}y)^2}{2}} (2\pi)^{-\frac{1}{2}} e^{-\frac{y^2}{2}} dy. \quad (93)$$

Expanding the inner exponent: $-\frac{e^{-2t}x^2 + 2e^{-t}\sqrt{1 - e^{-2t}}xy + (1 - e^{-2t})y^2}{2} - \frac{y^2}{2}$. Combine the y^2 terms: $-((1 - e^{-2t}) + 1)\frac{y^2}{2} = -(2 - e^{-2t})\frac{y^2}{2}$. The Gaussian integral in y then gives

$$(P_t f)(x) = (2 - e^{-2t})^{-\frac{1}{2}} e^{-e^{-2t}\frac{x^2}{2} + \frac{e^{-2t}(1 - e^{-2t})x^2}{2(2 - e^{-2t})}}. \quad (94)$$

After algebraic simplification, this reduces to

$$(P_t f)(x) = (2 - e^{-2t})^{-\frac{1}{2}} \exp\left(-\frac{x^2}{2(2 - e^{-2t})}\right). \quad (95)$$

(One can verify: at $t = 0$ this is $1 \cdot e^{-\frac{x^2}{2}} = f(x)$ ✓; as $t \rightarrow \infty$ it tends to $2^{-\frac{1}{2}}e^{-\frac{x^2}{4}}$, the equilibrium projection.)

Solution to Exercise 3.5. Eigenvalue -3 on $\mathbb{R}^2$ has multiplicity 4 (multi-indices α with $|\alpha| = 3$ in dimension 2): $\alpha = (3, 0), (2, 1), (1, 2), (0, 3)$. Eigenfunctions:

$$\begin{aligned} \text{He}_3(x_1) &= x_1^3 - 3x_1, & \text{He}_2(x_1)\text{He}_1(x_2) &= (x_1^2 - 1)x_2, \\ \text{He}_1(x_1)\text{He}_2(x_2) &= x_1(x_2^2 - 1), & \text{He}_3(x_2) &= x_2^3 - 3x_2. \end{aligned} \quad (96)$$

Orthogonality of $\text{He}_3(x_1)$ and $\text{He}_2(x_1)\text{He}_1(x_2)$: $\int \text{He}_3(x_1)(x_1^2 - 1)x_2 d\gamma_2 = [\int \text{He}_3(x_1^2 - 1) d\gamma_1] \cdot [\int x_2 d\gamma_1] = (\text{anything}) \cdot 0 = 0$. ✓

6.4 Solutions for §5.4

Solution to Exercise 4.1. On $\mathbb{C}^2 \simeq \mathbb{R}^4$ with $V = \|z\|^2 = a_1^2 + b_1^2 + a_2^2 + b_2^2$, $\nabla V = 2(a_1, b_1, a_2, b_2)$. Thus

$$\Delta_\mu f = \sum_{i=1}^2 (\partial_{a_i}^2 + \partial_{b_i}^2) f - 2(a_1 \partial_{a_1} + b_1 \partial_{b_1} + a_2 \partial_{a_2} + b_2 \partial_{b_2}) f. \quad (97)$$

For $f = a_1 a_2 - b_1 b_2$: all second partials are zero, and $2(a_1 \partial_{a_1} + \dots) f = 2(a_1 a_2 + b_1 b_2 + a_1 a_2 + b_1 b_2 - (\text{redo carefully}))$. Actually $a_1 \partial_{a_1}(a_1 a_2 - b_1 b_2) = a_1 a_2$; $b_1 \partial_{b_1} f = -b_1 b_2$; $a_2 \partial_{a_2} f = a_1 a_2$; $b_2 \partial_{b_2} f = -b_1 b_2$. Sum $= 2(a_1 a_2 - b_1 b_2) = 2f$. Hence $\Delta_\mu f = 0 - 2 \cdot 2f = -4f$. Wait, this has $V = \|z\|^2$ not $\frac{\|z\|^2}{2}$, so the eigenvalue scaling differs. With $V = \frac{\|z\|^2}{2}$, the OU operator's eigenvalues are $-2|\alpha|$ (a factor 2 from $\nabla V = 2z$ rather than z). The bilinear monomial $a_1 a_2$ has total degree $|\alpha| = 2$, so eigenvalue $-2 \cdot 2 = -4$. ✓

Solution to Exercise 4.2. The Fubini–Study metric on $\mathbb{CP}^1 \simeq S^2$ is $\frac{1}{4}$ of the round metric of radius 1 (by direct computation: pull back $g_{FS} = \frac{dw d\bar{w}}{(1+|w|^2)^2}$ via stereographic projection). Hence the Laplacian is $4\Delta_{S^2}$, with eigenvalues $-4\ell(\ell+1)$ for $\ell = 0, 1, 2, \dots$, i.e. $0, -8, -24, -48, \dots$

Solution to Exercise 4.3. On $\mathbb{CP}^2$ with affine chart (w_1, w_2) representing $[1 : w_1 : w_2]$, the energy expectation is $E([z]) = \frac{0 \cdot 1 + 1 \cdot |w_1|^2 + 2 \cdot |w_2|^2}{1 + |w_1|^2 + |w_2|^2}$. Setting $\partial_{w_j} E = 0$ gives the critical points: at $[1 : 0 : 0]$, $[0 : 1 : 0]$, $[0 : 0 : 1]$ (the eigenstates), E takes values $0, 1, 2$. The Hessian at $[1 : 0 : 0]$ in the chart (w_1, w_2) : to leading order, $E \approx |w_1|^2 + 2|w_2|^2$, positive definite, Morse index 0. At $[0 : 1 : 0]$ use the chart $(\zeta_0, \zeta_2) = (\frac{1}{w_1}, \frac{w_2}{w_1})$; expand E around $\zeta_0 = \zeta_2 = 0$: leading order is $-|\zeta_0|^2 + |\zeta_2|^2$, giving Morse index 2 (one negative complex direction = two negative real directions). At $[0 : 0 : 1]$: Morse index 4. ✓ Matches Atiyah–Bott prediction $2(k-1)$ for $k = 1, 2, 3$.

Solution to Exercise 4.4. As $\beta \rightarrow \infty$, $e^{-\beta E}$ concentrates near the minimum of E , which is at $[1 : 0 : 0]$ with $E = 0$. The Laplace-method estimate gives $\mu_\beta(B(x_0, r)) \approx e^{-\beta E_{\min}} \int_B e^{-\beta(E-E_{\min})} d\text{vol}_{FS}$, which by Gaussian asymptotics yields concentration at rate β^{-2} (the dimension of the local quadratic model is $4 = \text{real dimension of } \mathbb{CP}^2$, so the volume scales as β^{-2}). Total mass: $\mu_\beta(\mathbb{CP}^2) \sim C\beta^{-2}e^{-\beta E_{\min}}$.

Solution to Exercise 4.5. In the chart (w_1, w_2) , the inverse Fubini–Study metric is $g^{j\bar{\ell}} = (1 + \|w\|^2)(\delta_{j\bar{\ell}} + w_j \bar{w}_{\bar{\ell}})$ (by inverting the formula in §4.5). The energy function E in the chart is as in Exercise 4.3. The weighted Laplacian is

$$\Delta_{\mu_\beta} f = \Delta_{FS} f - \beta g_{FS}(\nabla E, \nabla f), \quad (98)$$

where Δ_{FS} is the unweighted Fubini–Study Laplacian. In coordinates this is a fourth-degree-rational PDE in w_1, w_2 ; the explicit form is most useful when expanded around a critical point of E for spectral computations. The leading-order quadratic part is what governs low-temperature thermal fluctuations.

Solution to Exercise 4.6. Embed $\iota : \mathbb{CP}^1 \times \mathbb{CP}^1 \rightarrow \mathbb{CP}^3$ via $([a], [b]) \mapsto [a \otimes b]$. In affine charts on $\mathbb{CP}^1$ with coordinates α, β (for the two factors), the embedding is

$$\iota(\alpha, \beta) = [1 \otimes 1 : 1 \otimes \beta : \alpha \otimes 1 : \alpha \otimes \beta] = [1 : \beta : \alpha : \alpha\beta]. \quad (99)$$

Pull back the Fubini–Study Kähler potential $K_{FS} = \log(1 + \sum_k |w_k|^2)$:

$$\begin{aligned} \iota^* K_{FS} &= \log(1 + |\beta|^2 + |\alpha|^2 + |\alpha\beta|^2) \\ &= \log((1 + |\alpha|^2)(1 + |\beta|^2)) \\ &= \log(1 + |\alpha|^2) + \log(1 + |\beta|^2). \end{aligned} \quad (100)$$

Since $\omega_{FS} = i\partial\bar{\partial}K_{FS}$, the pull-back is the **sum** of the two factor Fubini–Study forms:

$$\iota^* \omega_{FS} = \omega_{FS}^{(\alpha)} + \omega_{FS}^{(\beta)}, \quad (101)$$

which is exactly the product Kähler form on $\mathbb{CP}^1 \times \mathbb{CP}^1$. ✓

End of coursebook.

7 Notation summary

For ease of reference, we collect all notation used in the book.

Symbol	Meaning
M, N	smooth manifolds
$T_p M, T_p^* M$	tangent and cotangent spaces at p
g	Riemannian metric
g_{ij}, g^{ij}	metric in coordinates and its inverse
$d\text{vol}_g$	Riemannian volume form
∇f	gradient: $g(\nabla f, X) = X(f)$
$\text{div}_g X$	Riemannian divergence
Δ_g	Laplace–Beltrami operator: $\Delta_g f = \text{div}_g(\nabla f)$
V	potential function on M
$D = e^{-V}$	density
$d\mu = D d\text{vol}_g$	weighted measure
$\Omega_\mu = e^{-V} d\text{vol}_g$	weighted volume form
$\text{div}_\mu X$	weighted divergence: $\text{div}_g X - g(\nabla V, X)$
Δ_μ	weighted Laplacian: $\Delta_g f - g(\nabla V, \nabla f)$
Δ_W	Witten Laplacian
$d\gamma_n$	standard Gaussian measure on $\mathbb{R}^n$
Δ_γ	Ornstein–Uhlenbeck operator: $\Delta - x \cdot \nabla$
He_n	probabilist Hermite polynomial of degree n
$P_t = e^{t\Delta_\gamma}$	Ornstein–Uhlenbeck semigroup
$\mathcal{L}_X$	Lie derivative along X
ι_X	interior product / contraction with X
ω_{FS}	Fubini–Study form
g_{FS}	Fubini–Study (Hermitian) metric
$\mathcal{S}_n = \{z \in \mathbb{C}^n : \ z\ = 1\}$	unit sphere in $\mathbb{C}^n$
$\mathbb{CP}^{n-1} = \frac{\mathcal{S}_n}{S^1}$	complex projective space